Electronic Components

Q1 : Explain active and passive component :

- **Active Components**

These are components that can inject energy into a circuit and are capable of amplifying signals or controlling current flow. They require an external power source to function.

- Characteristics:
  1. Energy Control: Can control the flow of current or voltage.
  2. Amplification: Capable of amplifying signals (e.g., transistors).
  3. Dependency on Power Source: Operate only when powered by an external source.
- Examples:
  Diodes (in some configurations like LEDs or Zener diodes).
  Transistors (BJTs, MOSFETs).
  Integrated Circuits (ICs)

- **Passive Components**

These components cannot introduce energy into a circuit but can store, dissipate, or control energy in various ways. They do not require an external power source to function.

- Characteristics:
  1. Energy Consumption/Storage: Consume or store energy but do not amplify.
  2. Independent Operation: Do not require an external power source to function.
  3. Bidirectional Operation: Typically allow current to flow in both directions.

- Examples:
  Resistors (dissipate energy).
  Capacitors (store energy in the form of an electric field).
  Inductors (store energy in the form of a magnetic field).

- Key Differences:

  Understanding these distinctions is essential for designing and analyzing electrical and electronic circuits effectively.

Q2 Explain concept of resistor with diagram

- **Concept of a Resistor:**
  A resistor is a passive electronic component that resists the flow of electric current in a circuit. It is primarily used to control the current, divide voltage, or protect components from excessive current. The resistance of a resistor is measured in ohms (Ω)

- **Basic Principles:**

1. Ohm's Law:

The relationship between voltage (), current (), and resistance () is given by:

V = I.R

= Voltage across the resistor (in volts).

= Current through the resistor (in amperes).

= Resistance (in ohms).

2. Function:

Resistors limit current flow.

They dissipate electrical energy as heat.

Used in circuits to achieve specific voltage drops.

Symbol of Resistor:



- **Applications:**

1. Current Limiting: Protects components like LEDs from excessive current.
2. Voltage Division: Used in voltage divider circuits.
3. Signal Conditioning: Adjusts signal levels in electronic circuits.
4. Heat Generation: Used in heating elements.

Q3 Explain color coding of registers.

- **Resistor Color Code:**
  The color coding of resistors is a standardized way to indicate their resistance value, tolerance, and sometimes reliability or temperature coefficient. Resistors have colored bands printed on them, and each band corresponds to a specific value or multiplier.

- **Color Coding of Register:**

Color	Color	1st Band	2nd Band	3rd Band Multiplier	4th Band Tolerance
Black		0	0	x1Ω	
Brown		1	1	x10Ω	±1%
Red		2	2	x100Ω	±2%
Orange		3	3	x1kΩ	
Yellow		4	4	x10kΩ	
Green		5	5	x100kΩ	±0.5%
Blue		6	6	x1MΩ	±0.25%
Violet		7	7	x10MΩ	±0.10%
Grey		8	8	x100MΩ	±0.05%
White		9	9	x1GΩ	
Gold				x0.1Ω	±5%
Silver				x0.01Ω	±10%

- **Standard Resistor Bands:**
  - **4-Band Resistors:** Most common, with 4 bands indicating resistance and tolerance.
  - **5-Band Resistors:** Used for more precise values.

- **Example (4-Band Resistor):**
  Colors: Red, Violet, Yellow, Gold

1. Red = 2, Violet = 7 → Significant digits = 27
2. Yellow = → Multiplier = Ω or 270 kΩ
3. Gold = ±5% → Tolerance = 27000 ± 5

- **Applications:**
  - ✓ Used in circuits to select the appropriate resistor value for desired current or voltage control.
  - ✓ Helps in designing precise electronic components where accuracy is critical.

Q4Explain concept of capacitor.

- **Concept of a Capacitor:**

A capacitor is a passive electronic component that stores electrical energy in an electric field. It is widely used in electrical and electronic circuits for various purposes, including energy storage, filtering, timing, and coupling.

- **Structure:**

A capacitor consists of:

1. Two Conductive Plates: Made of metal or conductive material.

2. Dielectric Material: An insulating layer between the plates (e.g., air, ceramic, plastic, mica).

When voltage is applied, one plate accumulates positive charge, while the other accumulates an equal amount of negative charge.

- **Key Characteristics:**

1. Capacitance ():

The ability of a capacitor to store charge.

Measured in Farads (F).

Formula:

C = Q/V

- C = Capacitance.
- Q = Charge stored (in coulombs).
- V = Voltage across the capacitor (in volts).

- **Types of Capacitors:**

1. Fixed Capacitors: Have a constant capacitance.
Ceramic, electrolytic, film, mica capacitors.
2. Variable Capacitors: Capacitance can be adjusted.

- **Applications:**

1. Energy Storage: Temporarily stores charge in power supplies.

2. Filtering: Removes noise or smoothens voltage in circuits.

3. Timing Circuits: Determines the timing in oscillators or clocks.

4. Coupling/Decoupling: Passes AC signals while blocking DC.

- **Symbol of a Capacitor:**

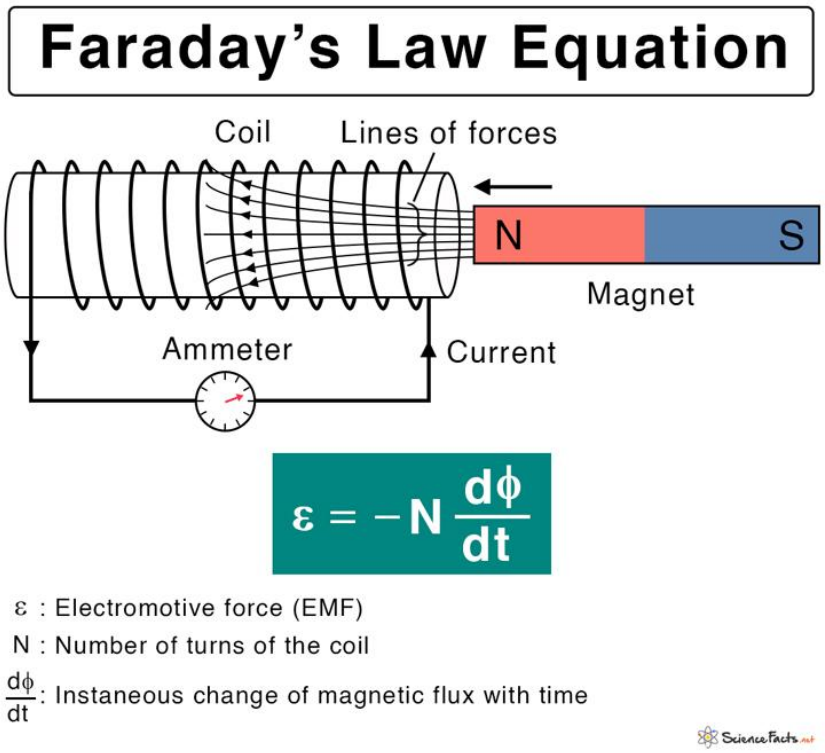
1. Fixed Capacitor:

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2. Polarized Capacitor (Electrolytic):

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Q5 Explain what is faraday's law of electromagnetic induction.



The discovery and understanding of electromagnetic induction are based on a long series of experiments carried out by Faraday and Henry. From the experimental observations, Faraday concluded that an emf is induced when the magnetic flux across the coil changes with time. Therefore, Faraday's first law of electromagnetic induction states the following:

Whenever a conductor is placed in a varying magnetic field, an electromotive force is induced. If the conductor circuit is closed, a current is induced, which is called induced current.

Q6 Give difference between resistor and capacitor and inductor.

Relation	Resistor (R)	Inductor (L)	Capacitor(C)
Voltage-Current	V=iR	$v = L\frac{di}{dt}$	$v = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$
Current-Voltage	$i = \frac{v}{R}$	$i = \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau + i(t_0)$	$i = C\frac{dv}{dt}$
Power & Energy	$p = i^2R = \frac{v^2}{R}$	$w = \frac{1}{2}Li^2$	$w = \frac{1}{2}Cv^2$
Series	$R_{eq} = R_1 + R_2$	$L_{eq} = L_1 + L_2$	$C_{eq} = \frac{C_1C_2}{C_1 + C_2}$
Parallel	$R_{eq} = \frac{R_1.R_2}{R_1 + R_2}$	$L_{eq} = \frac{L_1L_2}{L_1 + L_2}$	$C_{eq} = C_1 + C_2$
DC Excitation	Same	Short Circuit (SC)	Open circuit (OC)
Circuit Variable that cannot change abruptly	-	i	v

Q7 Compare DC Power supply with AC Power supply.

Alternating Current	Direct Current
You can transmit AC electrical energy over long distances without significant power losses.	DC loses electrical power if you transmit it over extensive distances.
It features a changing magnetic field which, in turn, changes the electric flow direction.	It has a stable magnetism. Hence current travels unidirectionally.
AC frequency is different from one country to the other. But, on average, most nations use 50 Hz or 60 Hz frequencies.	It features no frequency since it's steady.
There is a constant forward and backward periodic shifting of AC flow direction as it travels.	Contrastingly, DC power steadily travels in one order throughout.
Lastly, the electrons will move forward and backward during AC transmission.	The electrons move steadily in one direction like a stream of water.